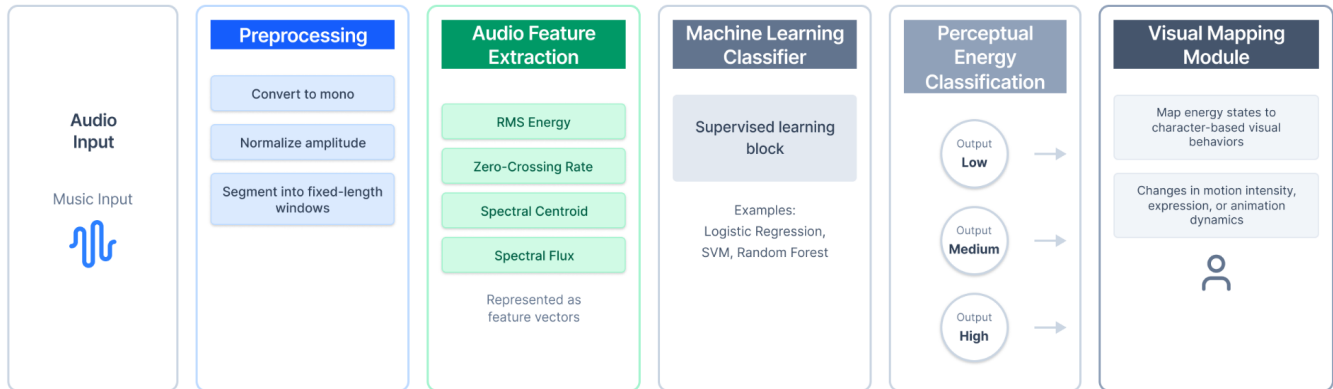


Music-to-Imagine System Team 18

1. Motivation and Problem Statement

As students without engineering or tech backgrounds, this project is our first hands-on attempt to build a system using signal processing and machine learning methods. We choose a manageable music-based system to help us understand how these techniques work on real-world signals.

Use audio feature extraction (e.g., RMS energy, spectral features) to analyze music signals and apply machine learning to classify perceptual energy states, which are then mapped to character-based visual behaviors.



2. Proposed Methodology

Audio signals will be converted to mono, normalized, and segmented into fixed-length windows (*Müller, 2015*). We will extract perceptually relevant audio features including RMS energy, zero-crossing rate, spectral centroid, and spectral flux, which are commonly used to represent perceived energy and activity in music (*Tzanetakis & Cook, 2002*). These features will be used to train supervised classifiers (e.g., Logistic Regression, SVM, or Random Forest) to predict perceptual energy states (Low / Medium / High), following standard audio classification pipelines (*Bishop, 2006*). Short audio clips will be manually labeled based on perceived energy, and model performance will be evaluated using accuracy and confusion matrices (*Powers, 2011*).

3. Expected Outcomes and Deliverables

Our final outcome can be an interactive music-to-imagination demo in which audio energy states are mapped to predefined visual outputs. Project success can be evaluated by basic classification results and the clarity of the visual expression of different energy states.

4. Challenges and Milestones

Subjective labeling and class overlap may affect classification, which will be handled using consistent labeling rules and simple feature-model comparisons. Limited dataset size will be addressed by keeping the task small and prioritizing a complete pipeline over high accuracy.

Project Milestones Gantt Chart



5. References

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